USER MANUAL

PROJECT 2: Colon Cancer Medical Diagnosis Simulator

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GROUP 6

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Introduction

Welcome to the Colorectal Cancer Diagnosis Simulator. This is a web-based decision-support tool that helps explore the risk of colorectal cancer in a patient. It combines three engines: a Machine Learning model trained on clinical data, a literature-based risk calculator (NCI CCRAT-style), and a deep learning model that detects suspicious tumor regions in CT images.

This manual explains how to use the application step by step. You do not need any technical background to follow it.

> **Important notice:** This application does NOT replace professional clinical judgment. Results are orientative and based on academic models that have not been clinically validated. Any decision regarding screening, diagnosis or treatment must be made by qualified medical staff.

First time you open the application

When the application loads, you see a centered title at the top with the project name, and four navigation tabs below. The tabs are:

1. Clinical risk: the main tab. Here you enter patient data and obtain a combined risk assessment.
2. CT image: upload a CT slice and see suspicious tumor regions highlighted.
3. Statistical analysis: charts from the complementary R analysis.
4. Information: documentation, scientific sources and glossary.

We recommend starting with the Information tab if it is your first time, to understand the system context. Then move to Clinical risk to start using the simulator.

Tab 1: Clinical risk

How to enter patient data

The form has 13 fields organized in two columns. Fill them in this order:

1. Age: use the slider (range 18 to 90 years).
2. Gender: select Male or Female.
3. Family history of colorectal cancer in first-degree relatives: Yes or No.
4. Smoking habit: Yes or No.
5. Regular alcohol consumption: Yes or No.
6. BMI category: Normal, Overweight or Obese.
7. Diet quality: Low risk, Moderate risk or High risk.
8. Physical activity: Low, Moderate or High.
9. Type 2 diabetes: Yes or No.
10. Inflammatory bowel disease (IBD): Yes or No.
11. Known genetic mutation (Lynch, FAP, MUTYH): Yes or No.
12. Residential environment: Urban or Rural.
13. Healthcare access: Good or Limited.

Once all fields are filled, press the "Evaluate clinical risk" button at the bottom.

Understanding the results

The application shows the results in three sections, top to bottom:

Section 1 — Per-engine analysis. Two cards side by side:

- Left card: Machine Learning model. Shows the predicted probability and a risk level badge.
- Right card: Literature calculator. Shows the absolute 5-year risk and a risk level badge.
- Each card has an expander "How is this value calculated?" with the technical details.

Section 2 — Combined decision:

- Final risk level: Low, Medium or High.
- Numeric score (0 to 100%).
- Horizontal gauge with green-yellow-red gradient.
- Step-by-step explanation: weights applied, clinical criteria triggered, discrepancies detected.
- Clinical recommendation adapted to the level.

Section 3 — Identified factors. Colored chips:

- Red chips: risk factors found (with their RR multiplier).
- Green chips: protective factors found.

Risk levels explained

Level	Score range	Meaning	Recommendation
Low	< 25%	Standard population risk	Standard screening from age 45-50
Medium	25-55%	Above-average risk	Periodic monitoring and lifestyle changes
High	> 55% or override	Significantly elevated risk	Referral to gastroenterology

Tab 2: CT image

How to upload an image

14. Click "Browse files" in the file uploader.
15. Select a CT slice file (.npz, .npy, .png, .jpg or .jpeg).
16. Wait 1-3 seconds for processing.

Reading the result

Two images appear side by side:

- Left: the original CT slice after preprocessing.
- Right: the same slice with suspicious regions highlighted in red.

Below: the estimated tumor probability and a recommendation. The expander "How does the model detect the tumor?" explains the U-Net workflow in plain language.

> **Caution:** The red region is the model prediction, not absolute truth. It always requires confirmation by a qualified radiologist. The model can produce both false positives (marking healthy tissue) and false negatives (missing real tumors).

Tab 3: Statistical analysis

This tab shows the complementary R analysis charts, organized in three sections:

Exploratory analysis (EDA)

- Age distribution by cohort (cases vs controls).
- Frequency of risk factors in each cohort.
- Variable correlogram showing relationships among clinical features.

Logistic regression with Odds Ratios

- Forest plot of OR with 95% confidence intervals.
- Comparison OR (model) vs RR (literature). Critical chart: shows where the model agrees with literature and where it disagrees.
- Forest plot of literature RR for reference.

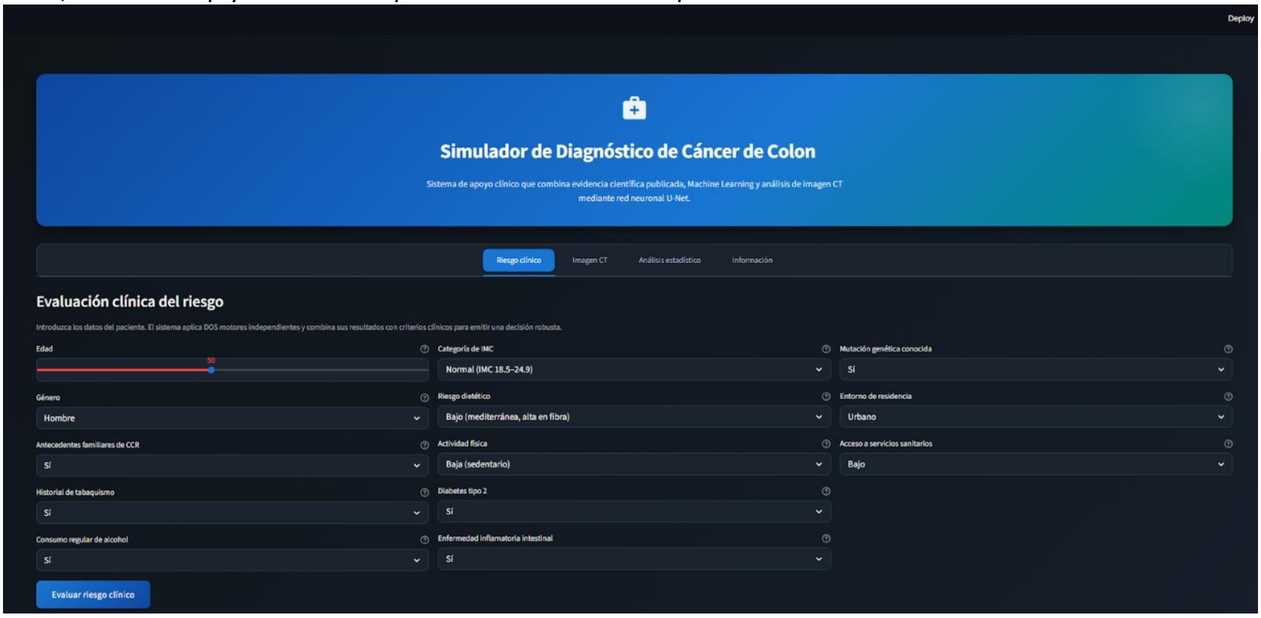
Model validation

- ROC curve with bootstrap confidence bands.
- Calibration plot.
- Prediction distribution by cohort.
- Sensitivity, specificity, precision and F1 by threshold.

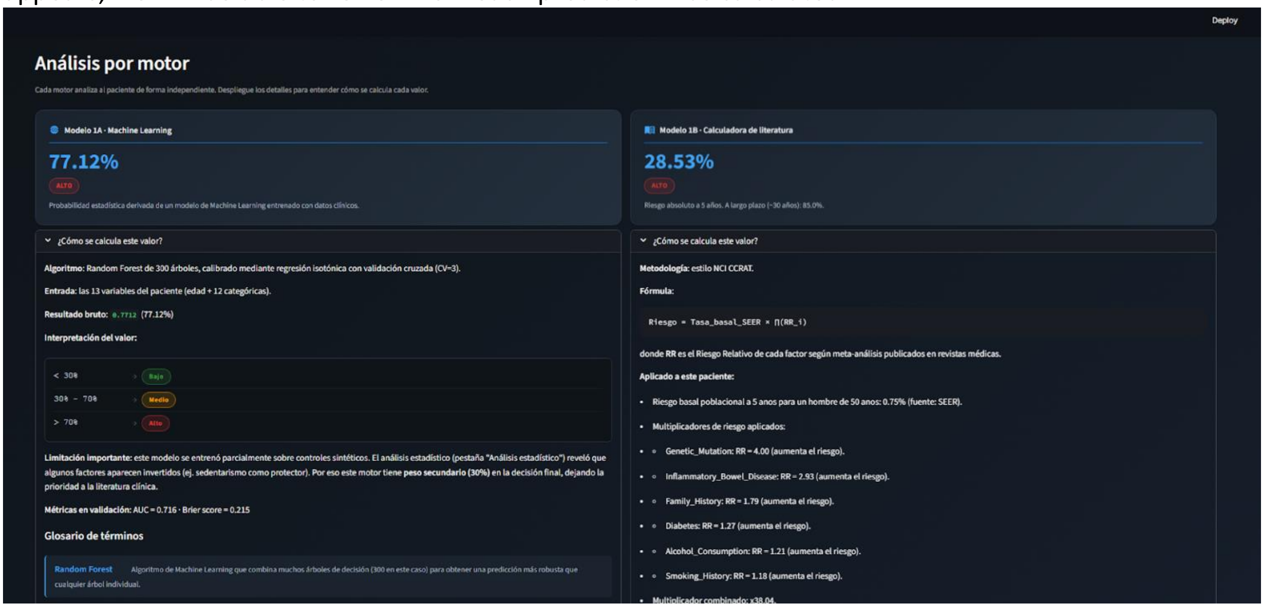
Why this tab matters: This is the most important tab to understand the academic value of the project. It documents the model limitations honestly and justifies the dual-engine design.

How to use it

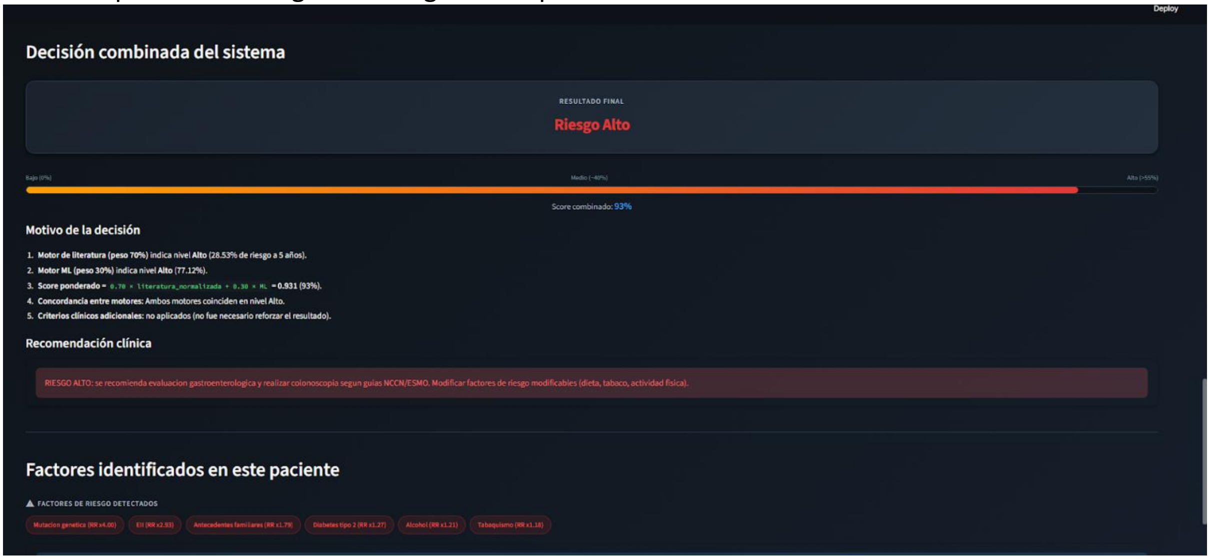
The first screen we will see is the prediction screen, where we will focus on analyzing the patient's medical characteristics in order to make a prediction and determine whether it is advisable to perform the CT analysis.
Here, we will simply fill in the required data and click on predict:



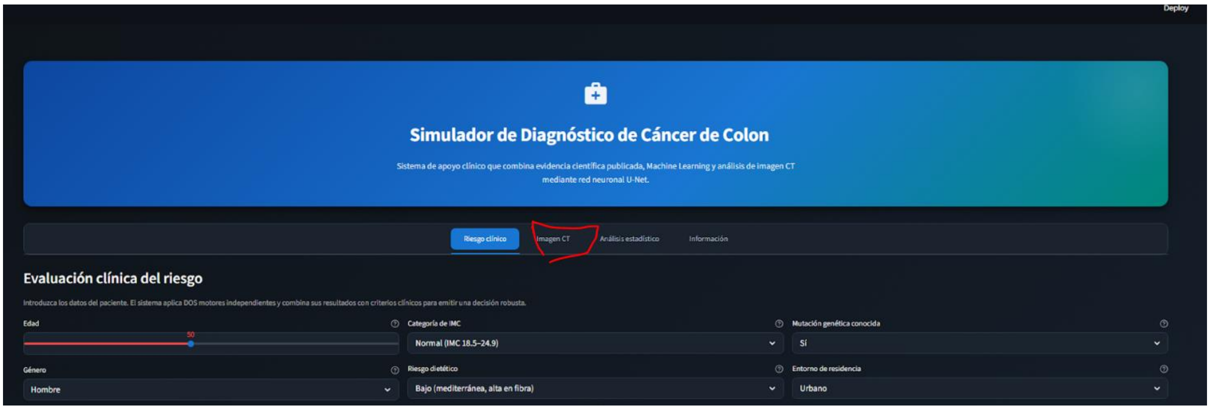
This will run a calculation using two different models to provide a final verdict. Once the result appears, we will be able to review how each prediction was calculated.



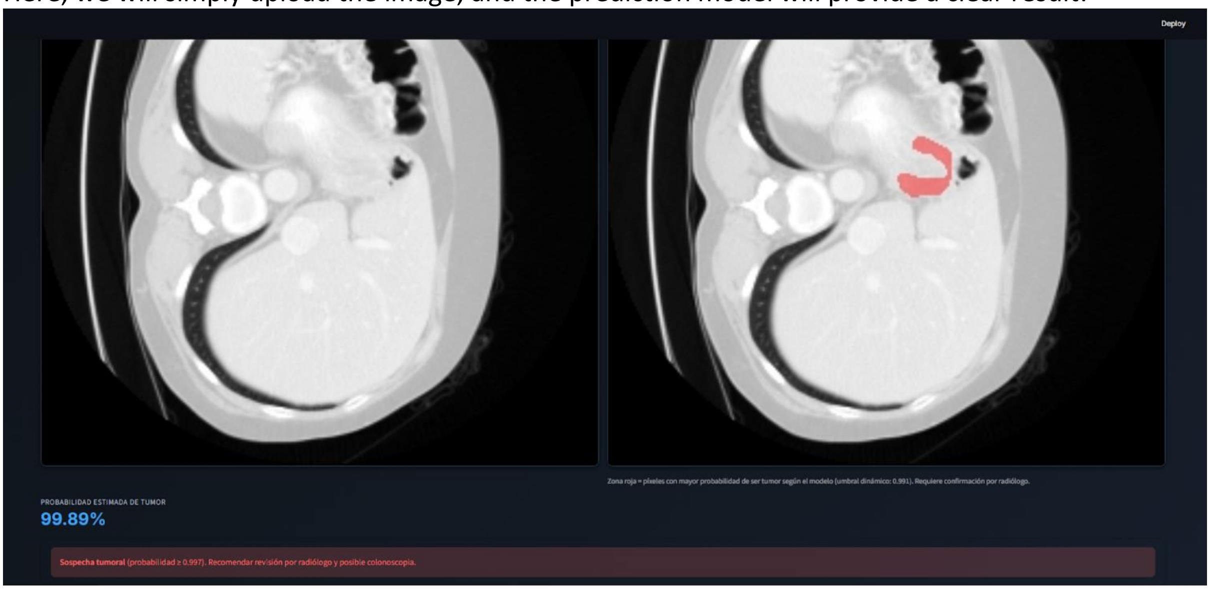
Using both results, a final recommendation will be provided, where it may be necessary to perform a scan and provide its images for image-based prediction.



If the result is high, we will need to access the image analysis screen to upload the scan images and perform the image-based prediction. To do this, on the main screen, we will go to the top section and select “CT Image.”



Here, we will simply upload the image, and the prediction model will provide a clear result:



Test cases you can try

Case 1: Healthy young patient

Input:

- Age 30, Female
- Family history: No, Smoking: No, Alcohol: No
- BMI: Normal, Diet: Low risk, Activity: High
- Diabetes: No, IBD: No, Mutation: No

Expected: Low risk, score < 10%, green gauge.

Case 2: Intermediate patient with modifiable factors

Input:

- Age 55, Male
- Family history: No, Smoking: Yes, Alcohol: Yes
- BMI: Obese, Diet: High risk, Activity: Low
- Diabetes: Yes, IBD: No, Mutation: No

Expected: Medium risk, additional clinical criterion activated due to 5+ modifiable factors.

Case 3: High risk with critical criterion

Input:

- Age 65, Male
- Family history: Yes, Smoking: Yes, Alcohol: Yes
- BMI: Obese, Diet: High risk, Activity: Low
- Diabetes: Yes, IBD: Yes, Mutation: Yes

Expected: High risk, override applied due to mutation + IBD, level forced to High regardless of numeric score.

Case 4: CT image validation

Use the example file data/processed/msd_colon_slices_256/colon_001_z052.npz (real patient slice with annotated tumor). The model should return a probability >99% and correctly highlight the tumor region.

Frequently asked questions

Can I use this for a real patient diagnosis?

No. This system is an academic prototype and has NOT been clinically validated. Any real medical decision must be made by qualified medical staff.

Why is the ML model only weighted 30%?

The ML model was trained partially on synthetic controls. The R statistical analysis (Tab 3) revealed that some coefficients are inverted compared to published literature. To mitigate this, the system gives more weight (70%) to the literature-based calculator, which is grounded in real cohorts and validated meta-analyses.

What does "discrepancy detected" mean?

When the ML model and the literature calculator give very different risk levels (e.g., one says Low and the other says High), the system shows a yellow warning banner. The system then prioritizes the literature decision because it is more reliable.

Why does mutation or IBD always force the level to High?

Because clinical guidelines (NCCN and ESMO) consider these factors as automatic indicators of high risk. The system applies these guidelines as override criteria regardless of the numeric score.

What if the CT model misses a tumor?

The CT model has a sensitivity of 73%, meaning it misses about 1 in 4 real tumors. This is why the application clearly states that the model output is orientative and always requires radiologist confirmation. Never rely on the CT result alone for clinical decisions.

Can I use my own CT image?

Yes, the application accepts .npz, .npy, .png, .jpg and .jpeg formats. The image will be normalized and resized automatically. Best results are obtained with axial slices similar to those in the training dataset (Medical Segmentation Decathlon).

How long does a prediction take?

Clinical prediction: less than 2 seconds on CPU. CT image: 2-3 seconds on CPU, less than 1 second on NVIDIA GPU.